

Practical Demo: Bootstrapping a Treasury Discount Curve

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This reading copy shows the calculations' results and figures. The HTML demo contains the full code in folded blocks and interactive scenario controls.

The desk question

A fixed-income analyst has Treasury prices. **What is today's value of each future dollar, and how can those values price a bond outside the input set?** This demo extends Lecture 4 with a different five-year dataset from payment schedules through curve construction, valuation, forward rates, and validation.

Purpose and assumptions

Sections 1–5 use ten **illustrative Treasury-like securities**, not market quotes. Face value is \$100, settlement is on a coupon date, accrued interest is zero, and payments align every six months. Rates are decimal annual nominal rates compounded semiannually unless stated otherwise. Default, options, taxes, liquidity premiums, and transaction costs are outside this teaching model.

Section 6 separately uses a **January 2, 2025 Treasury par-yield snapshot**. That historical dataset demonstrates the alternative route of starting with a published par curve. It is neither live data nor a file of individual bond prices.

1 Input Prices and Payment Schedules

$$P_i^{\text{full}} = P_i^{\text{clean}} + AI_i = \sum_j CF_{ij} D(t_j), \quad D(0) = 1.$$

Each row of the cash-flow matrix is one instrument. Each column is a payment date. The final payment includes principal. Contractual coupon rates and YTMs are different concepts: a YTM summarizes a bond's own price and payments.

Maturity (years)	Coupon (%)	Full price
0.500	0.000	98.100
1.000	3.500	99.550
1.500	4.000	100.100
2.000	4.000	99.750
2.500	4.250	100.350
3.000	4.500	101.000
3.500	4.500	100.800
4.000	4.750	101.350
4.500	5.000	102.100
5.000	5.125	102.400

Cash flows at payment dates 1–5

Maturity	0.5Y	1Y	1.5Y	2Y	2.5Y
0.5000	100.0000	0.0000	0.0000	0.0000	0.0000
1.0000	1.7500	101.7500	0.0000	0.0000	0.0000
1.5000	2.0000	2.0000	102.0000	0.0000	0.0000
2.0000	2.0000	2.0000	2.0000	102.0000	0.0000
2.5000	2.1250	2.1250	2.1250	2.1250	102.1250
3.0000	2.2500	2.2500	2.2500	2.2500	2.2500
3.5000	2.2500	2.2500	2.2500	2.2500	2.2500
4.0000	2.3750	2.3750	2.3750	2.3750	2.3750
4.5000	2.5000	2.5000	2.5000	2.5000	2.5000
5.0000	2.5625	2.5625	2.5625	2.5625	2.5625

Cash flows at payment dates 6–10

Maturity	3Y	3.5Y	4Y	4.5Y	5Y
0.5000	0.0000	0.0000	0.0000	0.0000	0.0000
1.0000	0.0000	0.0000	0.0000	0.0000	0.0000
1.5000	0.0000	0.0000	0.0000	0.0000	0.0000
2.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2.5000	0.0000	0.0000	0.0000	0.0000	0.0000
3.0000	102.2500	0.0000	0.0000	0.0000	0.0000
3.5000	2.2500	102.2500	0.0000	0.0000	0.0000
4.0000	2.3750	2.3750	102.3750	0.0000	0.0000
4.5000	2.5000	2.5000	2.5000	102.5000	0.0000
5.0000	2.5625	2.5625	2.5625	2.5625	102.5625

The six-month input pays only \$100. The one-year bond pays \$1.75 in six months and \$101.75 in one year. We must value each payment at its own date.

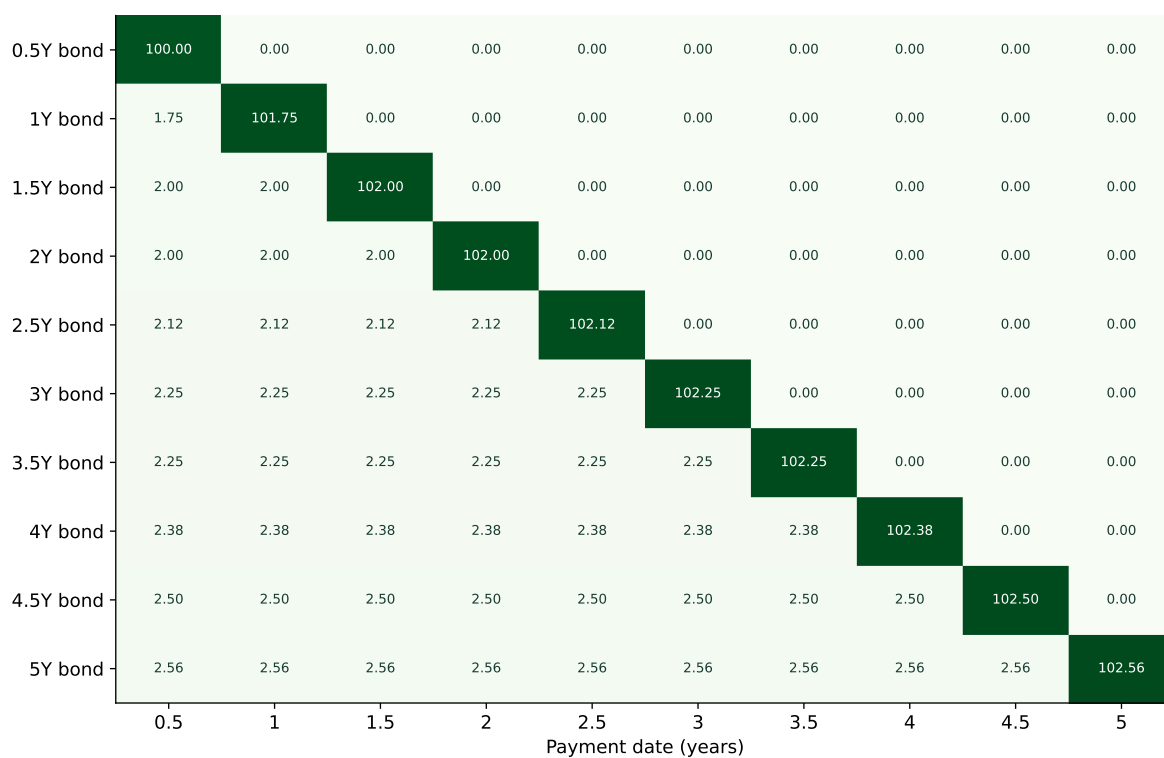


Figure 1: Dated payments in dollars per \$100 face value.

2 Bootstrapping the Discount Factors

For the bond maturing at $t_n = n/2$, let C_n be the semiannual coupon in dollars:

$$D(t_n) = \frac{P_n - C_n \sum_{j=1}^{n-1} D(t_j)}{100 + C_n}, \quad z(t_n) = 2 [D(t_n)^{-1/n} - 1] .$$

Subtract the value of the earlier coupons, whose discount factors are known. The residual price belongs to the final payment. Dividing by that payment gives its discount factor. The recursion needs one new unknown per instrument.

2.1 The first three steps by hand

$$D(0.5) = 98.10/100 = 0.981000.$$

$$D(1) = \frac{99.55 - 1.75(0.981000)}{101.75} = 0.961506143.$$

$$D(1.5) = \frac{100.10 - 2[0.981000 + 0.961506143]}{102} = 0.943284193.$$

$$z(1) = 2[D(1)^{-1/2} - 1] = 3.964208\%.$$

The remaining seven securities use the same recursion through year 5. The tables below retain unrounded factors and show every step.

2.2 The complete calculation trail

Years	Earlier PV	Residual price	Final payment	Discount factor
0.500000	0.000000	98.100000	100.000000	0.981000
1.000000	1.716750	97.833250	101.750000	0.961506
1.500000	3.885012	96.214988	102.000000	0.943284
2.000000	5.771581	93.978419	102.000000	0.921357
2.500000	8.090188	92.259812	102.125000	0.903401
3.000000	10.598734	90.401266	102.250000	0.884120
3.500000	12.588003	88.211997	102.250000	0.862709
4.000000	15.336271	86.013729	102.375000	0.840183
4.500000	18.243900	83.856100	102.500000	0.818108
5.000000	20.796400	81.603600	102.562500	0.795648

Years	Discount factor	Spot (%)
0.500000	0.981000	3.873598
1.000000	0.961506	3.964208
1.500000	0.943284	3.930637
2.000000	0.921357	4.137600
2.500000	0.903401	4.105119
3.000000	0.884120	4.147843
3.500000	0.862709	4.264189
4.000000	0.840183	4.401116
4.500000	0.818108	4.511477
5.000000	0.795648	4.624638

The spot curve and discount-factor curve contain the same information. The 1.5-year spot is below the one-year spot: a valid curve need not have monotonically increasing spot rates.

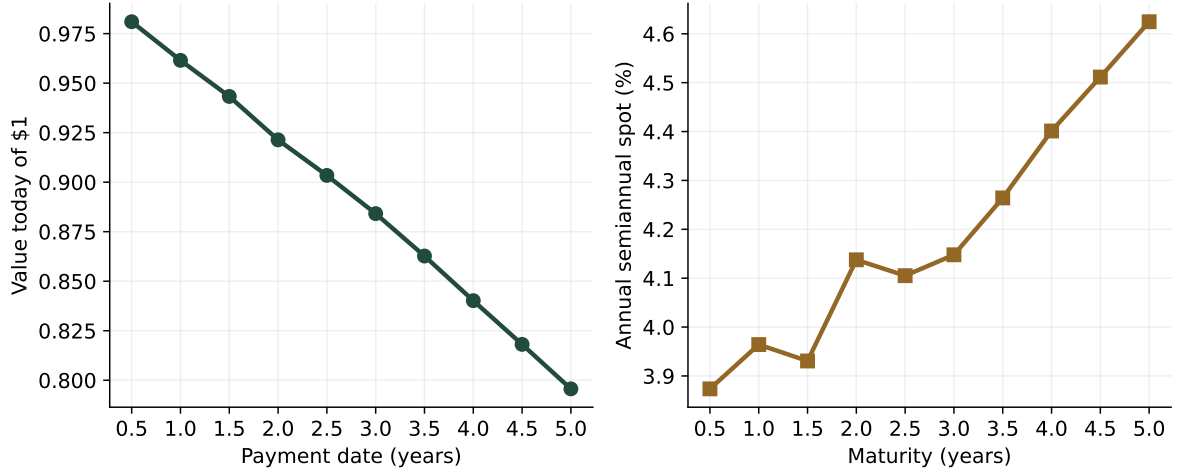


Figure 2: Two equivalent representations of the same pricing curve.

3 Repricing and Valuing a New Bond

$$\hat{P}_i = \sum_j CF_{ij}D(t_j), \quad e_i = \hat{P}_i - P_i.$$

Years	Input price	Repriced	Error per 100
0.50000000	98.10000000	98.10000000	0.00000000
1.00000000	99.55000000	99.55000000	0.00000000
1.50000000	100.10000000	100.10000000	0.00000000
2.00000000	99.75000000	99.75000000	0.00000000
2.50000000	100.35000000	100.35000000	0.00000000
3.00000000	101.00000000	101.00000000	0.00000000
3.50000000	100.80000000	100.80000000	0.00000000
4.00000000	101.35000000	101.35000000	0.00000000
4.50000000	102.10000000	102.10000000	0.00000000
5.00000000	102.40000000	102.40000000	0.00000000

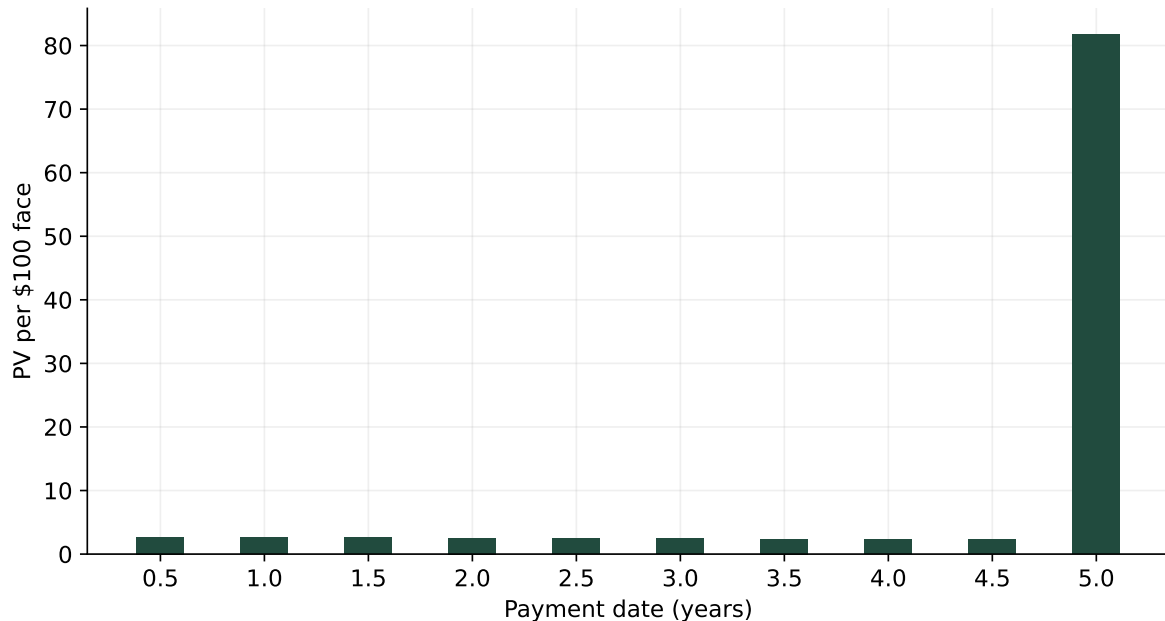
Near-zero residuals confirm the recursion. They do not establish that the input prices are synchronized, liquid, or economically reasonable.

Now value a five-year **5.50% coupon bond**, absent from the input file:

$$P_{\text{new}} = 2.75 \sum_{j=1}^9 D(j/2) + 102.75D(5), \quad V = \frac{\text{face value}}{100} P_{\text{new}}.$$

Years	Payment	Discount factor	Present value
0.500000	2.750000	0.981000	2.697750
1.000000	2.750000	0.961506	2.644142
1.500000	2.750000	0.943284	2.594032
2.000000	2.750000	0.921357	2.533732
2.500000	2.750000	0.903401	2.484352
3.000000	2.750000	0.884120	2.431330
3.500000	2.750000	0.862709	2.372450
4.000000	2.750000	0.840183	2.310503
4.500000	2.750000	0.818108	2.249798
5.000000	102.750000	0.795648	81.752784

Full price: 104.070872 per 100. **Value of 2 million dollars face:** 2,081,417.43 dollars.



The final payment dominates because it includes principal. This valuation assumes cash flows comparable to those underlying the Treasury curve. Credit risk and embedded options require additional modeling.

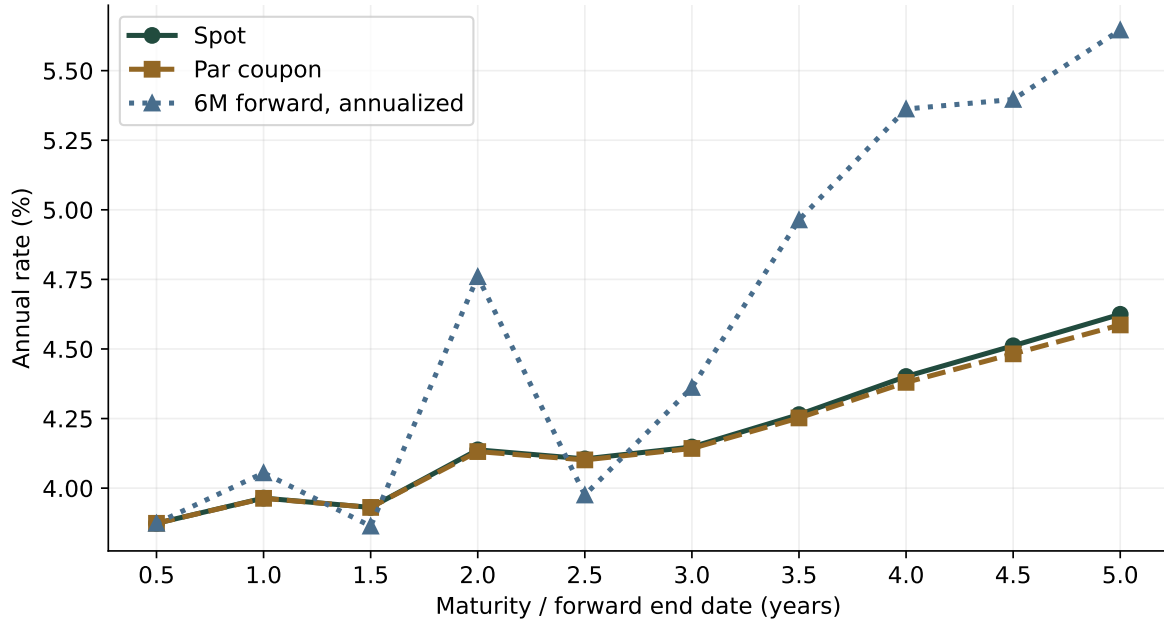
4 Derived Par Coupons and Forward Rates

The par coupon makes a hypothetical bond worth 100. A forward is the break-even rate between adjacent payment dates:

$$c_{\text{par}}(t_n) = \frac{2[1 - D(t_n)]}{\sum_{j=1}^n D(t_j)}, \quad f_{t_{n-1}, t_n}^{(2)} = 2 \left[\frac{D(t_{n-1})}{D(t_n)} - 1 \right].$$

The superscript (2) denotes annual nominal rates with semiannual compounding. The six-month interval return is $f^{(2)}/2$.

End date	Spot (%)	Par (%)	6M forward, annual (%)
0.500000	3.873598	3.873598	3.873598
1.000000	3.964208	3.963319	4.054859
1.500000	3.930637	3.930695	3.863512
2.000000	4.137600	4.131332	4.759749
2.500000	4.105119	4.101397	3.975246
3.000000	4.147843	4.142517	4.361599
3.500000	4.264189	4.252221	4.963656
4.000000	4.401116	4.380013	5.362181
4.500000	4.511477	4.482483	5.396512
5.000000	4.624638	4.586359	5.645907



The five-year par coupon is about **4.5864%**. The input bond pays 5.125%, so its price exceeds 100. The last forward is about **5.6459% annualized** and covers 4.5 to 5 years. It is a break-even rate, not a forecast.

4.1 Interactive quote sensitivity

Change only the five-year input price. Every setting rebuilds the curve. Earlier discount factors remain fixed, so this is **not a parallel rate shift**.

💡 Predict, change, explain

Predict what a higher five-year input price does to its discount factor, spot rate, final forward rate, and the new bond's value. Move the control, then explain why only the final payment's present value changes.

The table provides three settings of the same calculation, including a static reference for the PDF reading copy.

Price change	5Y spot (%)	Last forward (%)	New bond price
-1.000000	4.877097	8.197231	103.069044
0.000000	4.624638	5.645907	104.070872
1.000000	4.375559	3.156356	105.072700

5 Interpolation and Model Risk

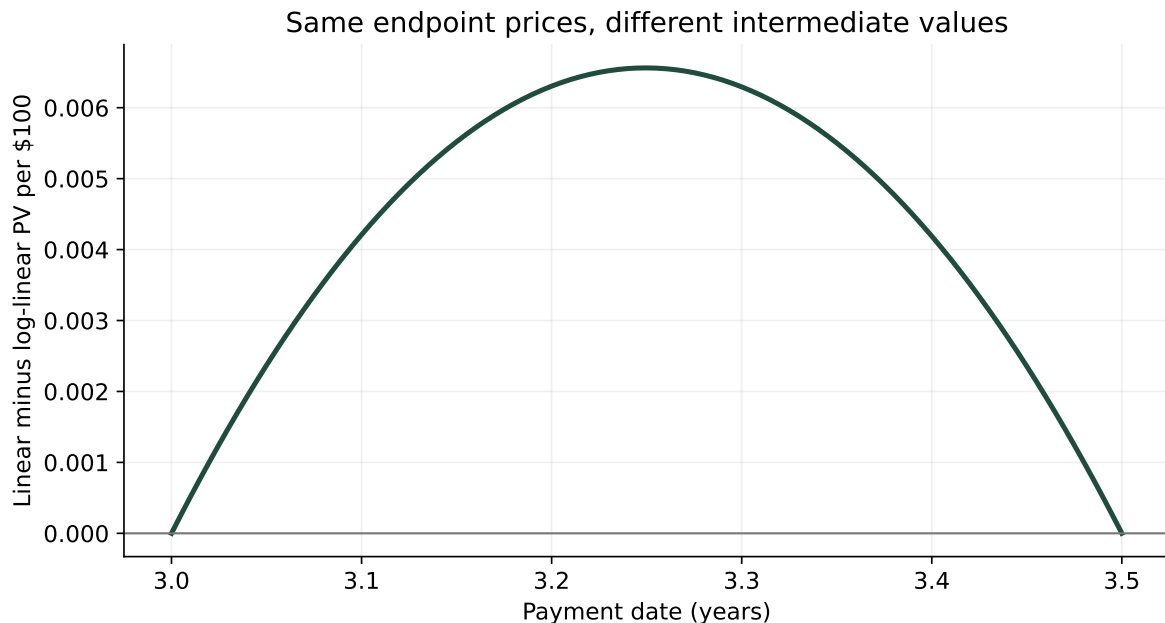
Suppose a claim pays \$100 at 3.25 years. There is no input at that date. Between known dates a, b , define $w = (t - a)/(b - a)$:

$$D_{\text{linear}}(t) = (1 - w)D(a) + wD(b),$$

$$D_{\text{log}}(t) = \exp\{(1 - w) \log D(a) + w \log D(b)\}.$$

Both methods reproduce the endpoint factors. Log-linear interpolation implies constant continuously compounded forwards within each interval. Neither intermediate value is observed, and neither rule guarantees smooth forwards at the knots.

Method	D(3.25)	PV of 100
Log-linear	0.87334888	87.33488778
Linear discounts	0.87341449	87.34144890



The difference is small over this short interval but not zero. Larger gaps and larger positions can make interpolation more important. An exact fit at known dates cannot establish the correct shape between them.

6 Market Data: An Existing Par Curve

These **January 2, 2025** observations come from [U.S. Treasury Daily Par Yield Curve Rates](#). They are published curve points, not security-level prices. The embedded snapshot makes the demonstration reproducible without an online data connection.

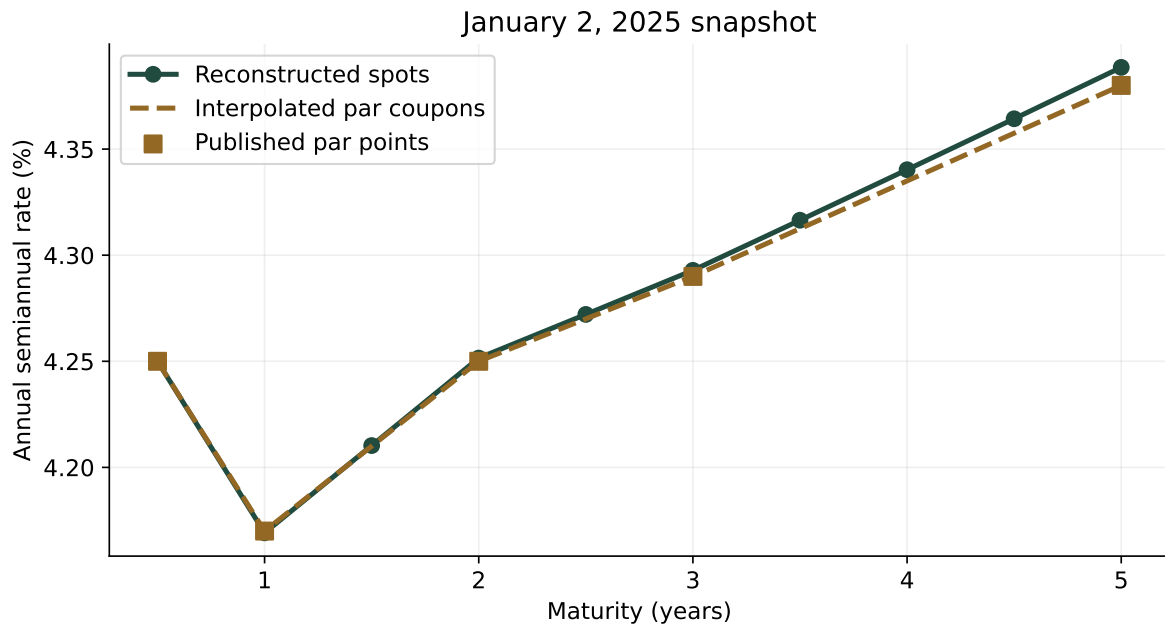
We interpolate the missing half-year par coupons linearly, then assign a price of 100 to each hypothetical par bond:

$$P_n = 100, \quad C_n = 100c_n/2, \quad D(t_n) = \frac{100 - C_n \sum_{j < n} D(t_j)}{100 + C_n}.$$

This is a **teaching reconstruction**, not the Treasury's exact internal curve. The [official methodology](#) uses its own market inputs and fitting procedure.

Date	Years	Published par (%)
2025-01-02	0.50	4.25
2025-01-02	1.00	4.17
2025-01-02	2.00	4.25
2025-01-02	3.00	4.29
2025-01-02	5.00	4.38

Years	Input origin	Par (%)	Spot (%)	D(t)
0.500000	Published	4.250000	4.250000	0.979192
1.000000	Published	4.170000	4.169166	0.959577
1.500000	Interpolated	4.210000	4.210287	0.939414
2.000000	Published	4.250000	4.251511	0.919303
2.500000	Interpolated	4.270000	4.272088	0.899715
3.000000	Published	4.290000	4.292853	0.880361
3.500000	Interpolated	4.312500	4.316470	0.861165
4.000000	Interpolated	4.335000	4.340278	0.842186
4.500000	Interpolated	4.357500	4.364286	0.823427
5.000000	Published	4.380000	4.388500	0.804888



The six-month par and spot quotes coincide because there is only one payment. At longer maturities, the par coupon generally differs from the terminal spot.

For live implementation, obtain synchronized security-level quotes, schedules, accrued interest, settlement and day-count conventions. Convert bill quotes correctly. An interpolated point is not a new market observation, and a dated snapshot does not represent today's market.

7 Validation Checks

The checks answer separate questions: does the recursion reproduce prices, does an independent matrix solution agree, do rate conventions reconcile, and do hypothetical par bonds return to 100?

All checks passed: input repricing, independent matrix solution, spot/forward identities, par reconstruction, all 41 quote scenarios, and both invalid-input tests.

Positive discount factors are essential. Decreasing factors hold in this positive-rate example but are not universal: negative rates can produce increasing factors. Extreme forward jumps warrant investigation of stale prices, inconsistent inputs or interpolation. A successful repricing check does not establish market-data quality.

8 Mini Project

Prepare a curve-validation memo for a portfolio manager. Use the ten illustrative inputs to value a five-year 5.50% coupon bond with \$2 million face. Then raise only the five-year calibration price by \$0.50 and repeat.

Deliver:

1. Input prices, cash-flow matrix, settlement and compounding assumptions.
2. Discount factors, spot rates, par coupons and forwards.
3. Input repricing errors and payment-by-payment valuation of the new bond.
4. Base-versus-shocked results, dollar P&L and an explanation.
5. The 3.25-year \$100 claim under both interpolation rules and limitations relevant to a live trade.

Worked deliverable: curve-validation memo

Decision summary. The baseline bond is worth 104.070872 per 100, or 2,081,417.43 dollars for 2 million face. Raising the five-year input price from 102.40 to 102.90 raises the position value to 2,091,435.72 dollars, a gain of 10,018.28 dollars. This is an instantaneous quote scenario, not a forecast.

Inputs and conventions. Prices are illustrative full prices per 100. Settlement is on a coupon date, accrued interest is zero, and semiannual cash flows share one payment grid. Rate quotes are annual nominal with semiannual compounding.

Maturity (years)	Coupon (%)	Full price
0.500	0.000	98.100
1.000	3.500	99.550
1.500	4.000	100.100
2.000	4.000	99.750
2.500	4.250	100.350
3.000	4.500	101.000
3.500	4.500	100.800
4.000	4.750	101.350
4.500	5.000	102.100
5.000	5.125	102.400

Cash flows at payment dates 1–5

Maturity	0.5Y	1Y	1.5Y	2Y	2.5Y
0.5000	100.0000	0.0000	0.0000	0.0000	0.0000
1.0000	1.7500	101.7500	0.0000	0.0000	0.0000
1.5000	2.0000	2.0000	102.0000	0.0000	0.0000
2.0000	2.0000	2.0000	2.0000	102.0000	0.0000
2.5000	2.1250	2.1250	2.1250	2.1250	102.1250
3.0000	2.2500	2.2500	2.2500	2.2500	2.2500
3.5000	2.2500	2.2500	2.2500	2.2500	2.2500
4.0000	2.3750	2.3750	2.3750	2.3750	2.3750
4.5000	2.5000	2.5000	2.5000	2.5000	2.5000
5.0000	2.5625	2.5625	2.5625	2.5625	2.5625

Cash flows at payment dates 6–10

Maturity	3Y	3.5Y	4Y	4.5Y	5Y
0.5000	0.0000	0.0000	0.0000	0.0000	0.0000
1.0000	0.0000	0.0000	0.0000	0.0000	0.0000
1.5000	0.0000	0.0000	0.0000	0.0000	0.0000
2.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2.5000	0.0000	0.0000	0.0000	0.0000	0.0000
3.0000	102.2500	0.0000	0.0000	0.0000	0.0000
3.5000	2.2500	102.2500	0.0000	0.0000	0.0000
4.0000	2.3750	2.3750	102.3750	0.0000	0.0000
4.5000	2.5000	2.5000	2.5000	102.5000	0.0000
5.0000	2.5625	2.5625	2.5625	2.5625	102.5625

Years	D(t)	Spot (%)	Par (%)	Forward (%)
0.500000	0.981000	3.873598	3.873598	3.873598
1.000000	0.961506	3.964208	3.963319	4.054859
1.500000	0.943284	3.930637	3.930695	3.863512
2.000000	0.921357	4.137600	4.131332	4.759749
2.500000	0.903401	4.105119	4.101397	3.975246
3.000000	0.884120	4.147843	4.142517	4.361599
3.500000	0.862709	4.264189	4.252221	4.963656
4.000000	0.840183	4.401116	4.380013	5.362181
4.500000	0.818108	4.511477	4.482483	5.396512
5.000000	0.795648	4.624638	4.586359	5.645907

Pricing control. Maximum absolute repricing error is 0.0000000000 per 100.

Years	Payment	D(t)	PV
0.500000	2.750000	0.981000	2.697750
1.000000	2.750000	0.961506	2.644142
1.500000	2.750000	0.943284	2.594032
2.000000	2.750000	0.921357	2.533732
2.500000	2.750000	0.903401	2.484352
3.000000	2.750000	0.884120	2.431330
3.500000	2.750000	0.862709	2.372450
4.000000	2.750000	0.840183	2.310503
4.500000	2.750000	0.818108	2.249798
5.000000	102.750000	0.795648	81.752784

Measure	Base	Input price +0.50
D(5)	0.795648	0.800523
5Y spot (%)	4.624638	4.499681
Last forward (%)	5.645907	4.393551
New bond price	104.070872	104.571786

Calculation and interpretation. Earlier discounts stay fixed. The final factor increases by $0.50/102.5625 = 0.004875076$. Only the new bond's final payment changes in present value: $\Delta P = 102.75(0.50/102.5625) = 0.500914077$ per 100. Multiply by 20,000 to obtain the position gain. The higher discount factor corresponds to a lower spot and a lower final forward rate.

Interpolation. The 3.25-year claim is worth 87.33488778 under log-linear interpolation and 87.34144890 under linear discount interpolation. Both reproduce the endpoints. Their difference is model risk, not evidence of executable arbitrage.

Limitations and recommendation. The teaching model omits actual calendars, accrued interest, bid-ask costs, liquidity and financing effects. Before trading, validate synchronized quotes and schedules, inspect forward stability, and test interpolation sensitivity. The historical par snapshot is a separate example, not a substitute for these inputs.